# Supplementary material for *Dynamic consequences of declining vaccination coverage in a heterogeneous world*

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## 1 Theoretical honeymoon time in the SIR model

[**Emily:** *I've rewritten the first half of this section to define the model equations and derive the equilibria explicitly.*] We consider here how reductions in vaccination can lead to rebounds in disease prevalence. In particular, we will focus on the post-vaccination "honeymoon", or how long it takes for the disease to rebound after vaccination starts declining. We use the term "effective immunization coverage" throughout for the proportion successfully immunized, thereby accounting for both vaccination coverage and imperfect vaccine efficacy.

We begin with an SIR (Susceptible, Infected, Recovered) model of disease transmission. We assume that births and deaths occur at the same rate, $\mu$, and some proportion, $v$, of births are effectively immunized. We further assume that new infections occur upon contact between susceptible and infected individuals at rate $\beta$, and infected individuals recover at rate $\gamma$. The model can be defined by the following system of ordinary differential equations:

$$\frac{dS}{dt} = \mu(1-v) - \beta SI - \mu S \tag{1}$$

$$\frac{dI}{dt} = \beta SI - (\gamma + \mu)I \tag{2}$$

$$\frac{dR}{dt} = \mu v + \gamma I - \mu R \tag{3}$$

where $S$, $I$, and $R$ represent the proportions of the population that are susceptible, infected, and recovered, respectively.

One way to understand the dynamics of the SIR model is through its equilibrium behavior, or the conditions under which the system does not change over time. The equilibrium conditions are given by setting the derivatives to zero, and for the $I$ compartment, we have

$$\frac{dI}{dt} = \beta SI - (\gamma + \mu)I = I(\beta S - (\gamma + \mu)) = 0, \tag{4}$$

which demonstrates the existence of two equilibria:

1. disease-free equilibrium, where $I^* = 0$ and $S^* = 1 - v$, and

2. endemic equilibrium, where $I^* = \frac{\mu}{(\gamma+\mu)}\big((1-v) - 1/\mathcal{R}_0\big) > 0$ and $S^* = \frac{\gamma+\mu}{\beta} = 1/\mathcal{R}_0$. This equilibrium point only exists and is positive when $v < 1 - 1/\mathcal{R}_0$. [**Emily:** *the exact equation for $I^*$ in this case is less relevant for our purposes, but I've included it here for completeness*]

Within this framework, we imagine a population with some "pre-decline" effective immunization coverage, $v_0$, that experiences an instantaneous drop in coverage to $v$. We consider only the disease-free case in which the pre-decline effective immunization coverage is above the herd-immunity threshold, $v_0 > 1 - 1/\mathcal{R}_0$, equivalently $S^* = 1 - v_0 < 1/\mathcal{R}_0$; otherwise there is no positive honeymoon period. [**Emily:** *flagging that we haven't formally introduced the definition of honeymoon period yet, but perhaps safe to say the reader has been through the main text and has the general idea*]

We hypothesize that the duration of the honeymoon period, or informally, the time until the first rebound outbreak, will depend on a number of factors including the basic reproduction number of the pathogen $\mathcal{R}_0$, the equilibrium proportion susceptible $S^*$, the birth rate $\mu$, the post-decline effective immunization coverage $v$, and the timing of importations. Because the timing and establishment of importations cannot be predicted by this deterministic model, we define the theoretical honeymoon time as the time until $\mathcal{R}_{\text{eff}} = \mathcal{R}_0 S(t)$ crosses 1. This provides a lower bound on the actual rebound time, because rebound additionally requires an importation to arrive and establish.

We can calculate analytically the time at which $\mathcal{R}_{\text{eff}}$ crosses 1 under the assumptions of local extinction ($I = 0$), constant post-decline effective immunization coverage $v$, and a constant population scaled to 1. The proportion susceptible satisfies

$$\frac{dS}{dt} = \mu(1 - v) - \mu S = \mu\big[(1 - v) - S\big]. \tag{5}$$

Because the population is initially at the disease-free equilibrium associated with the pre-decline effective immunization coverage $v_0$, we have $S(0) = 1 - v_0$. Since $v < v_0$, we have $S(0) < 1 - v$, so Equation (5) shows that $S(t)$ increases toward $1 - v$. Thus we have a first-order linear initial-value problem, whose solution is

$$S(t) = 1 - v - (v_0 - v)e^{-\mu t}. \tag{6}$$

The critical effective immunization coverage (corresponding to the herd-immunity threshold) is

$$v_{\text{crit}} = 1 - \frac{1}{\mathcal{R}_0}. \tag{7}$$

If effective immunization coverage declines from above to below this level, so that

$$v < v_{\text{crit}} < v_0, \tag{8}$$

then $S(t)$ increases from $1 - v_0$ and crosses $1/\mathcal{R}_0 = 1 - v_{\text{crit}}$ after a finite time. Setting $\mathcal{R}_{\text{eff}}(t) = \mathcal{R}_0 S(t) = 1$ gives the theoretical honeymoon time,

$$T = \frac{1}{\mu} \log\left(\frac{v_0 - v}{v_{\text{crit}} - v}\right) = \frac{1}{\mu} \log\left(\frac{v_0 - v}{1 - 1/\mathcal{R}_0 - v}\right). \tag{9}$$

This expression is positive because $v < v_{\text{crit}} < v_0$ implies $(v_0 - v)/(v_{\text{crit}} - v) > 1$. If $v \geq v_{\text{crit}}$, $\mathcal{R}_{\text{eff}}$ never exceeds 1 and the theoretical honeymoon time is infinite; if $v_0 \leq v_{\text{crit}}$,

the population is not initially above the herd-immunity threshold and there is no positive honeymoon period of this kind.

Equivalently, the pre- and post-decline values of effective immunization coverage that yield the same honeymoon time $T$ satisfy

$$v_0 = v + e^{\mu T}(v_{\text{crit}} - v) = v\big(1 - e^{\mu T}\big) + e^{\mu T} v_{\text{crit}}. \tag{10}$$

For fixed $T$ and $\mu$, Equation (10) defines a straight line in the plane whose horizontal and vertical coordinates are the post- and pre-decline effective immunization coverages, $v$ and $v_0$, respectively. The line has slope $1 - e^{\mu T} < 0$ and vertical intercept $e^{\mu T} v_{\text{crit}}$. Consequently, combinations of $v$ and $v_0$ that produce the same theoretical honeymoon time lie along straight lines. This explains the linear contours in the heatmap of theoretical honeymoon time (main text, Figure 2B).

# 2 Age-structured model

[**Emily:** *This section has been updated and all code/figures have been rerun. Minor numerical differences and no qualitative changes.*]

We also test honeymoon times under various assumptions about age-structured contact patterns. Our age-structured model defines a discrete set of $A$ age classes. Transmission between individuals of different age classes is governed by a who-acquires-infection-from-whom (WAIFW) matrix, $\theta_{i,j}$, where $i$ and $j$ index age classes. The full system of ordinary differential equations governing the proportion of individuals in each age class and disease state is as follows:

$$\frac{dS_{i=1}}{dt} = \mu(1 - v) - \sum_{j=1}^{A} \theta_{1,j} S_1 I_j - \mu S_1 - \alpha_1 S_1 \tag{11}$$

$$\frac{dS_{i\neq 1}}{dt} = \alpha_{i-1} S_{i-1} - \sum_{j=1}^{A} \theta_{i,j} S_i I_j - \mu S_i - \alpha_i S_i \tag{12}$$

$$\frac{dI_{i=1}}{dt} = \sum_{j=1}^{A} \theta_{1,j} S_1 I_j - (\gamma + \mu) I_1 - \alpha_1 I_1 \tag{13}$$

$$\frac{dI_{i\neq 1}}{dt} = \alpha_{i-1} I_{i-1} + \sum_{j=1}^{A} \theta_{i,j} S_i I_j - (\gamma + \mu) I_i - \alpha_i I_i \tag{14}$$

$$\frac{dR_{i=1}}{dt} = \mu v + \gamma I_1 - \mu R_1 - \alpha_1 R_1 \tag{15}$$

$$\frac{dR_{i\neq 1}}{dt} = \alpha_{i-1} R_{i-1} + \gamma I_i - \mu R_i - \alpha_i R_i \tag{16}$$

where $\mu$ is the birth and death rate, $\alpha_i$ is the rate of aging out of age class $i$ which is inversely proportional to its size, $\gamma$ is the recovery rate, and $v$ is the effective immunization coverage at birth. There is no aging out of the final age class to ensure a constant population scaled to one (i.e., $\alpha_A = 0$).

To calculate $\mathcal{R}_0$ in the age-structured model, we use the Next Generation Matrix method [1]. To do so, we follow the steps from Bjørnstad (2018) [2]:

1. Define the infected states, $\mathbf{X} = (I_1, I_2, ..., I_A)$

2. Define the new infection terms, $\mathcal{F}(\mathbf{X}) = \left[\sum_{j=1}^{A} \theta_{i,j} S_i I_j\right]$ for ages $i = 1, 2, ..., A$

3. Define all losses out of infected states,

$$\mathcal{V}^-(\mathbf{X}) = \begin{bmatrix} (\gamma + \mu + \alpha_1) I_1 \\ \vdots \\ (\gamma + \mu + \alpha_{A-1}) I_{A-1} \\ (\gamma + \mu) I_A \end{bmatrix}$$

4. Define all gained transfers between infected states,

$$\mathcal{V}^+(\mathbf{X}) = \begin{bmatrix} 0 \\ \alpha_1 I_1 \\ \vdots \\ \alpha_{A-1} I_{A-1} \end{bmatrix}$$

5. Define the change in infected states,

$$\mathcal{V}(\mathbf{X}) = \mathcal{V}^-(\mathbf{X}) - \mathcal{V}^+(\mathbf{X}) = \begin{bmatrix} (\gamma + \mu + \alpha_1) I_1 \\ (\gamma + \mu + \alpha_2) I_2 - \alpha_1 I_1 \\ \vdots \\ (\gamma + \mu + \alpha_{A-1}) I_{A-1} - \alpha_{A-2} I_{A-2} \\ (\gamma + \mu) I_A - \alpha_{A-1} I_{A-1} \end{bmatrix}$$

6. Compute the Jacobians of $\mathcal{F}$ and $\mathcal{V}$. The Jacobian of $\mathcal{V}$ is

$$\mathbf{V} = \left[\frac{\partial \mathcal{V}_i}{\partial X_j}\right] = \begin{bmatrix} \gamma + \mu + \alpha_1 & 0 & 0 & \cdots & 0 & 0 \\ -\alpha_1 & \gamma + \mu + \alpha_2 & 0 & \cdots & 0 & 0 \\ 0 & -\alpha_2 & \gamma + \mu + \alpha_3 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & -\alpha_{A-2} & \gamma + \mu + \alpha_{A-1} & 0 \\ 0 & 0 & \cdots & 0 & -\alpha_{A-1} & \gamma + \mu \end{bmatrix}$$

   and the Jacobian of $\mathcal{F}$ is

$$\mathbf{F} = \left[\frac{\partial \mathcal{F}_i}{\partial X_j}\right] = \begin{bmatrix} \theta_{1,1} S_1 & \theta_{1,2} S_1 & \cdots & \theta_{1,A} S_1 \\ \theta_{2,1} S_2 & \theta_{2,2} S_2 & \cdots & \theta_{2,A} S_2 \\ \vdots & \vdots & \ddots & \vdots \\ \theta_{A,1} S_A & \theta_{A,2} S_A & \cdots & \theta_{A,A} S_A \end{bmatrix}$$

7. Then, $\mathcal{R}_0$ is the dominant eigenvalue of the next generation matrix, $\mathbf{K} = \mathbf{F}\mathbf{V}^{-1}$ evaluated at the disease-free equilibrium for a particular vaccination level and assuming a stable age distribution. Similarly, $\mathcal{R}_{\text{eff}}(t)$ is the dominant eigenvalue evaluated at time $t$. We compute these values numerically.

# 3 The equivalence ratio

[**Emily:** *I've removed the "unity" terminology. The only reason to keep it would be that "unity calculation" is how Ottar describes it in his book, which I think is the (only? or) primary reference/derivation. "Equivalence ratio" is the term I coined for the modified version I created, but I'm open to other names. I suggest we settle on the approach and notation/terminology here, and then I will update the main text accordingly.*]

The dynamics of the age-structured model can be understood in relation to a model of aggregate variables where individuals are assumed to mix homogeneously (i.e., all susceptible individuals mix with all infective and all recovered individuals equally). We will refer to these two models as the age-structured model and the homogeneous model, respectively. To equate the dynamics under these two models, incidence can be written as

$$\begin{aligned} \text{age-structured infections} &= \sum_{i=1}^{A}\sum_{j=1}^{A}\theta_{i,j}S_i(t)I_j(t) \\ &= \hat{\beta(t)}\sum_{i=1}^{A}S_i(t)\sum_{j=1}^{A}I_j(t) \end{aligned} \tag{17}$$

given the time-varying transmission rate, $\hat{\beta(t)}$, defined as

$$\hat{\beta(t)} = \frac{\sum_{i=1}^{A}\sum_{j=1}^{A}\theta_{i,j}S_i(t)I_j(t)}{\sum_{i=1}^{A}S_i(t)\sum_{j=1}^{A}I_j(t)}. \tag{18}$$

We demonstrate this calculation for the age-structured model and example WAIFW matrices considered in the main text (Figure S1). For the flat WAIFW matrix, which is equivalent to the homogeneous model, $\hat{\beta(t)}$ is constant over time (flat black line, Figure S1A). In contrast, the other WAIFW matrices yield time-varying $\hat{\beta(t)}$, and when each $\hat{\beta(t)}$ is used in a homogeneous model of aggregate variables, the dynamics approximate those of the age-structured model (Figure S1B, solid colored versus gray dashed lines). In these simulation experiments, the infection dynamics in the two models agree to numerical precision (Figure reffig:unity-beta-simpleC).

We can also characterize the periods in which $\hat{\beta(t)}$ is higher or lower than the homogeneous case (flat black line, Figure S1A) via

$$\frac{\hat{\beta(t)}}{\beta} = \frac{\sum_{i=1}^{A}\sum_{j=1}^{A}\theta_{i,j}S_i(t)I_j(t)}{\beta\sum_{i=1}^{A}S_i(t)\sum_{j=1}^{A}I_j(t)}. \tag{19}$$

This ratio scales the homogeneous transmission rate, $\beta$, such that $\frac{\hat{\beta(t)}}{\beta} > 1$ indicates an increase in transmission over the homogeneous case, and vice versa for $\frac{\hat{\beta(t)}}{\beta} < 1$.

While the existence of $\hat{\beta(t)}$ indicates that the dynamics of the age-structured model can be reproduced by a homogeneous model with a time-varying transmission rate, it is not directly comparable across WAIFW matrices because the homogeneous model is reproducing varying age-structured dynamics. Instead, we can use the homogeneous case as a comparator, instead by rescaling transmission in the age-structured model:

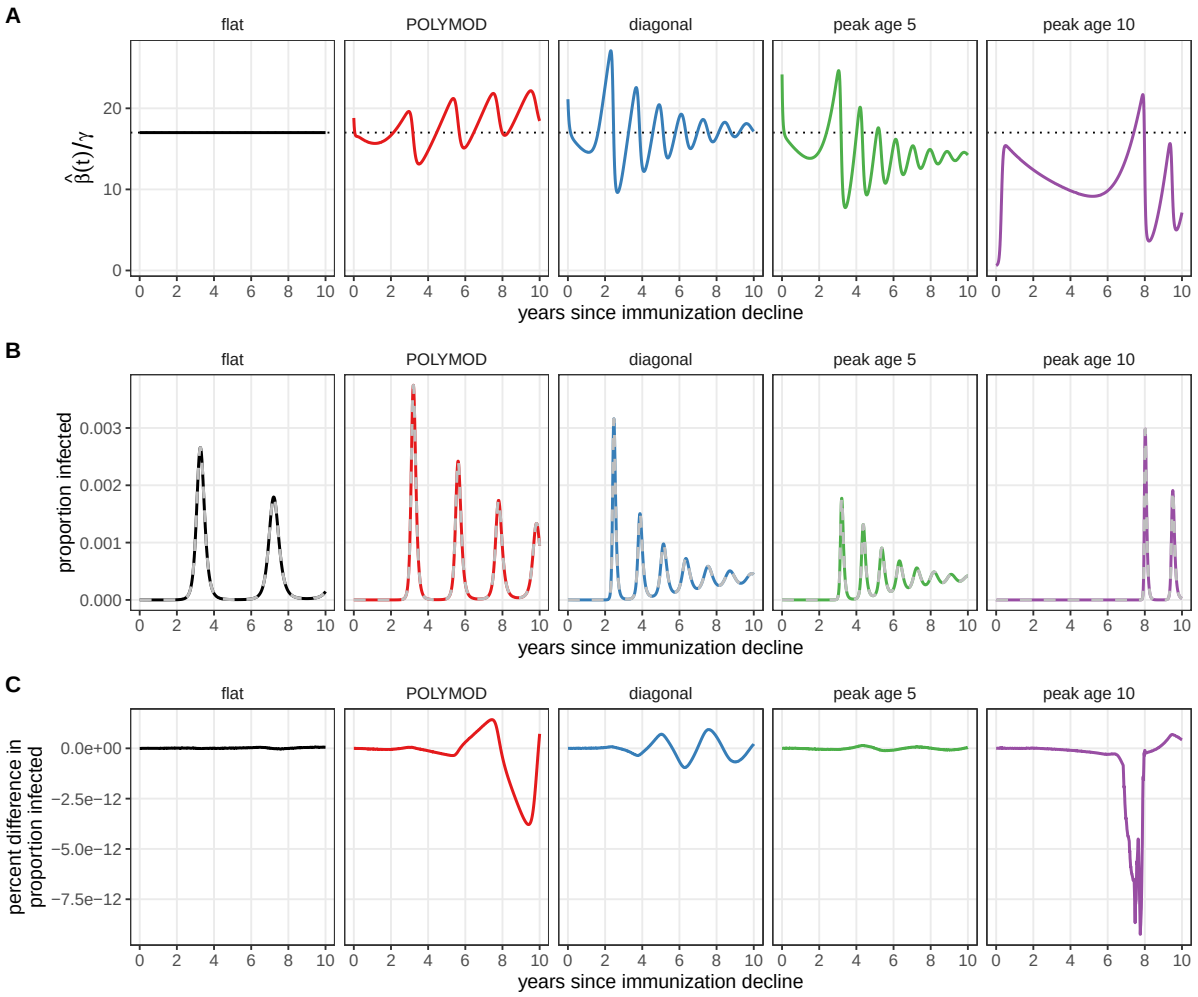



Figure S1: Illustration of $\hat{\beta(t)}/\gamma$ for five sample WAIFW matrices. (A) Time-varying transmission rate, $\hat{\beta(t)}$, scales homogeneous dynamics to age-structured. The flat line for the flat matrix is the annual transmission rate that yields $\mathcal{R}_0 = 17$ given $\gamma = 1/14$ days$^{-1}$ and $\mu = 1/80$ years$^{-1}$ (dotted line). (B) Infection dynamics since immunization decline in the age-structured model under different WAIFW matrices are shown in the solid colored lines (equivalent to Main text Figure 3), and the dashed gray lines (overlaid) show infection dynamics in a homogeneous model with time-varying transmission rate, $\hat{\beta(t)}$. (C) Percent difference in proportion of individuals infected between the age-structured model and homogeneous model with time-varying transmission rate, $\hat{\beta(t)}$. Percent difference is calculated as $\frac{\text{homogeneous infections} - \text{age-structured infections}}{\text{age-structured infections}}$. See Main text for details on sample WAIFW matrices.

$$\begin{aligned} \text{homogeneous infections} &= \beta \sum_{i=1}^{A} S_i(t) \sum_{j=1}^{A} I_j(t) \\ &= k(t) \sum_{i=1}^{A} \sum_{j=1}^{A} \theta_{i,j} S_i(t) I_j(t) \end{aligned} \tag{20}$$

where

$$k(t) = \frac{\beta \sum_{i=1}^{A} S_i(t) \sum_{j=1}^{A} I_j(t)}{\sum_{i=1}^{A} \sum_{j=1}^{A} \theta_{i,j} S_i(t) I_j(t)}. \tag{21}$$

In this case, $k(t)$ is a ratio scaling the age-structured transmission rates captured by $\theta_{i,j}$ and thus functions similarly to $\beta\hat{(t)}/\beta$. We call this quantity the equivalence ratio. Again, $k(t) > 1$ suggests that age-structured transmission increases over baseline to reproduce homogeneous dynamics, which implies that age-structured dynamics would have been slower than homogeneous in the absence of scaling.

[**Emily:** *I've chosen $k(t)$ notation to differentiate it from the $\hat{\beta}$ quantities above, but open to suggestions.*]

We again demonstrate the equivalence ratio calculation for the age-structured model and example WAIFW matrices (Figure S2). As with $\beta\hat{(t)}$, no scaling of transmission is necessary in the flat WAIFW case, and thus $k(t) = 1$ for all $t$. For the other WAIFW matrices, $k(t)$ varies over time, and when $k(t)$ scales the age-structured transmission rates, the dynamics match those of the homogeneous model (dashed lines in Figure S2B). The value of $k(t)$ relative to 1 suggests the degree to which the age-structured model deviates from the homogeneous model. For example, under the peak at age 10 WAIFW matrix, $k(t) > 1$ for the entire period, indicating that dynamics would have been slower under the age-structured assumptions than they would have been under the homogeneous model.

# 4 Metapopulation model

We consider a simple metapopulation model with $P$ patches that are assumed to be equally connected, such that $c$ is the proportion of contacts that occur within a patch and $1-c$ is the proportion of contacts that occur between patches. These between-patch contacts are assumed to be distributed uniformly across the remaining $P-1$ patches, and there is no permanent migration between patches. Patches are assumed to be identical except for a patch-specific effective immunization coverage, $v_p$. The model is defined by the following system of ordinary differential equations:

$$\frac{dS_p}{dt} = \mu(1 - v_p) - \beta \sum_{q=1}^{P} C_{p,q} S_p I_q - \mu S_p \tag{22}$$

$$\frac{dI_p}{dt} = \beta \sum_{q=1}^{P} C_{p,q} S_p I_q - (\gamma + \mu) I_p \tag{23}$$

$$\frac{dR_p}{dt} = \mu v_p + \gamma I_p - \mu R_p \tag{24}$$

where $\mathbf{C}$ is the connectivity matrix, with $C_{p,p} = c$ for all $p = 1, \cdots, P$ and $C_{p,q} = (1-c)/(P-1)$ for $p \neq q$. Again, we use the Next Generation Matrix method [1] to calculate $\mathcal{R}_0$ following the steps from Bjørnstad (2018) [2]:

1. Define the infected states, $\mathbf{X} = (I_1, I_2, ..., I_P)$
2. Define the new infection terms, $\mathcal{F}(\mathbf{X}) = \left[\beta \sum_{q=1}^{P} C_{p,q} S_p I_q\right]$ for patches $p = 1, 2, ..., P$

3. Define all losses out of infected states,

$$\mathcal{V}^{-}(\mathbf{X}) = \begin{bmatrix} (\gamma+\mu)I_1 \\ \vdots \\ (\gamma+\mu)I_P \end{bmatrix}$$

4. Define all gained transfers between infected states, $\mathcal{V}^{+}(\mathbf{X}) = 0$

5. Define the change in infected states,

$$\mathcal{V}(\mathbf{X}) = \mathcal{V}^{-}(\mathbf{X}) - \mathcal{V}^{+}(\mathbf{X}) = \begin{bmatrix} (\gamma+\mu)I_1 \\ \vdots \\ (\gamma+\mu)I_P \end{bmatrix}$$

6. Compute the Jacobians of $\mathcal{F}$ and $\mathcal{V}$. The Jacobian of $\mathcal{V}$ is a diagonal matrix, $\mathbf{V} = \text{diag}(\gamma+\mu)$, and thus the inverse $\mathbf{V}^{-1} = \text{diag}\left(\frac{1}{\gamma+\mu}\right)$. The Jacobian of $\mathcal{F}$ is

$$\mathbf{F} = \left[\frac{\partial \mathcal{F}_p}{\partial X_q}\right] = \beta \begin{bmatrix} C_{1,1}S_1 & C_{1,2}S_1 & \cdots & C_{1,P}S_1 \\ C_{2,1}S_2 & C_{2,2}S_2 & \cdots & C_{2,P}S_2 \\ \vdots & \vdots & \ddots & \vdots \\ C_{P,1}S_P & C_{P,2}S_P & \cdots & C_{P,P}S_P \end{bmatrix}$$

7. This yields the next generation matrix, $\mathbf{K} = \mathbf{F}\mathbf{V}^{-1} = \frac{\beta}{\gamma+\mu}\text{diag}(\mathbf{S})\mathbf{C}$ where $\mathbf{S} = (S_1, S_2, ..., S_P)$. We evaluate $\mathbf{K}$ at the disease-free equilibrium or at time $t$ and calculate the dominant eigenvalue numerically to obtain $\mathcal{R}_0$ and $\mathcal{R}_{\text{eff}}$, respectively.

# 5 Supplementary Figures

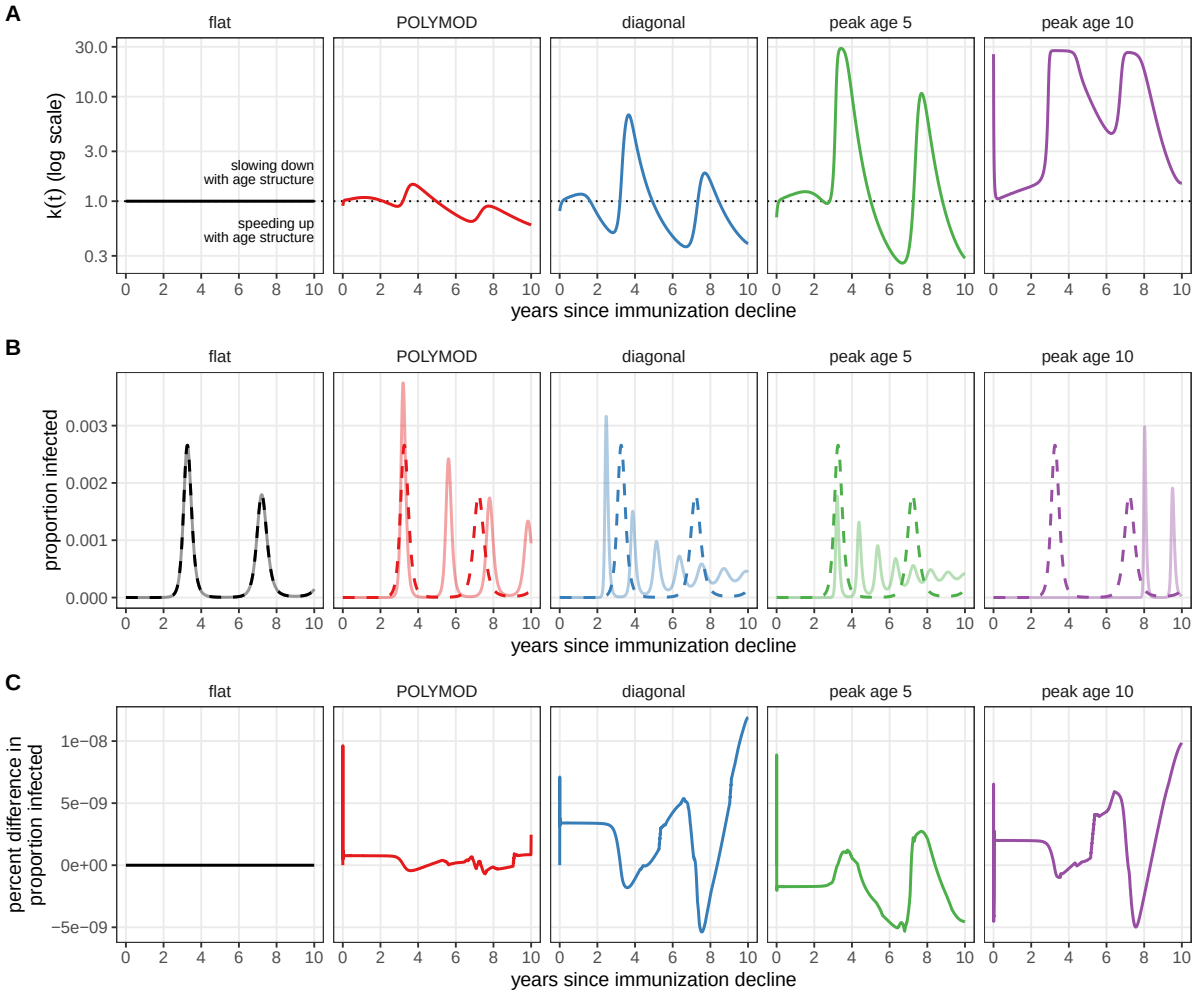



Figure S2: Illustration of $k(t)$ for five example WAIFW matrices. (A) Equivalence ratio, $k(t)$, scales age-structured transmission to yield dynamics equivalent to the homogeneous model. When $k(t) < 1$, unscaled transmission in the age-structured model is higher than the homogeneous model, indicating that age-structure would have "sped up" dynamics; vice versa when $k(t) > 1$. (B) Infection dynamics since immunization decline in the age-structured model under different WAIFW matrices are shown in the light solid lines (equivalent to Main text Figure 3), and dashed lines show infection dynamics when $k(t)$ is used to scale transmission in the age-structured model. See Main text for details on sample WAIFW matrices. (C) Percent difference in proportion of individuals infected between the homogeneous model and the age-structured model model with time-varying transmission rate, $\hat{\beta(t)}$. Percent difference is calculated as $\frac{\text{age-structured infections} - \text{homogeneous infections}}{\text{homogeneous infections}}$. See Main text for details on sample WAIFW matrices.

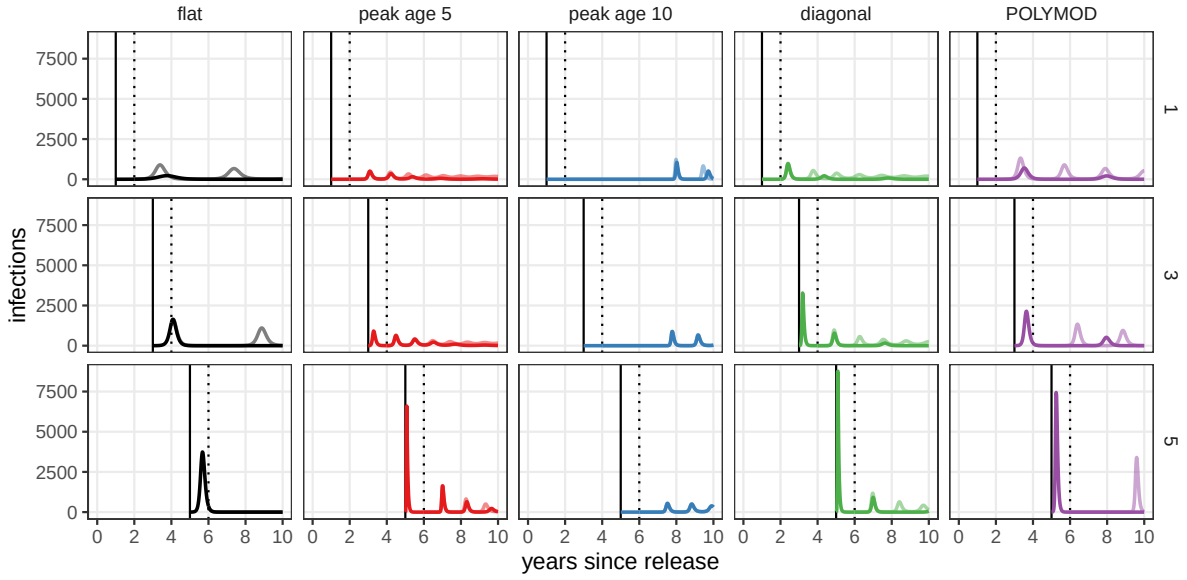



Figure S3: Sensitivity of results to the timing of infection introduction. For each WAIFW matrix (columns, see main text Figure 3A), simulated dynamics when a single infection is introduced into the 5 year old age class 1, 3, or 5 years after the drop in effective immunization coverage (rows). The solid vertical line shows the timing of introduction. The lighter line shows simulation results if effective immunization coverage returns to its pre-decline level 1 year after the drop (shown by dotted line).

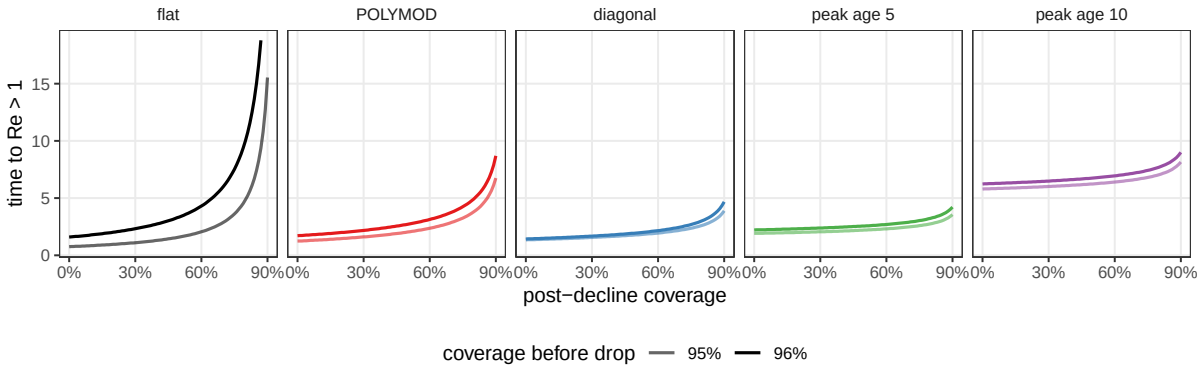



Figure S4: Theoretical honeymoon time predictions, computed as time until $\mathcal{R}_{\text{eff}} > 1$, under five assumptions about age-structured mixing. Results are shown for a population with $\mu = 1/80$ years$^{-1}$, a pathogen with $\mathcal{R}_0 = 17$, and pre-decline effective immunization coverage of 95% and 96%.

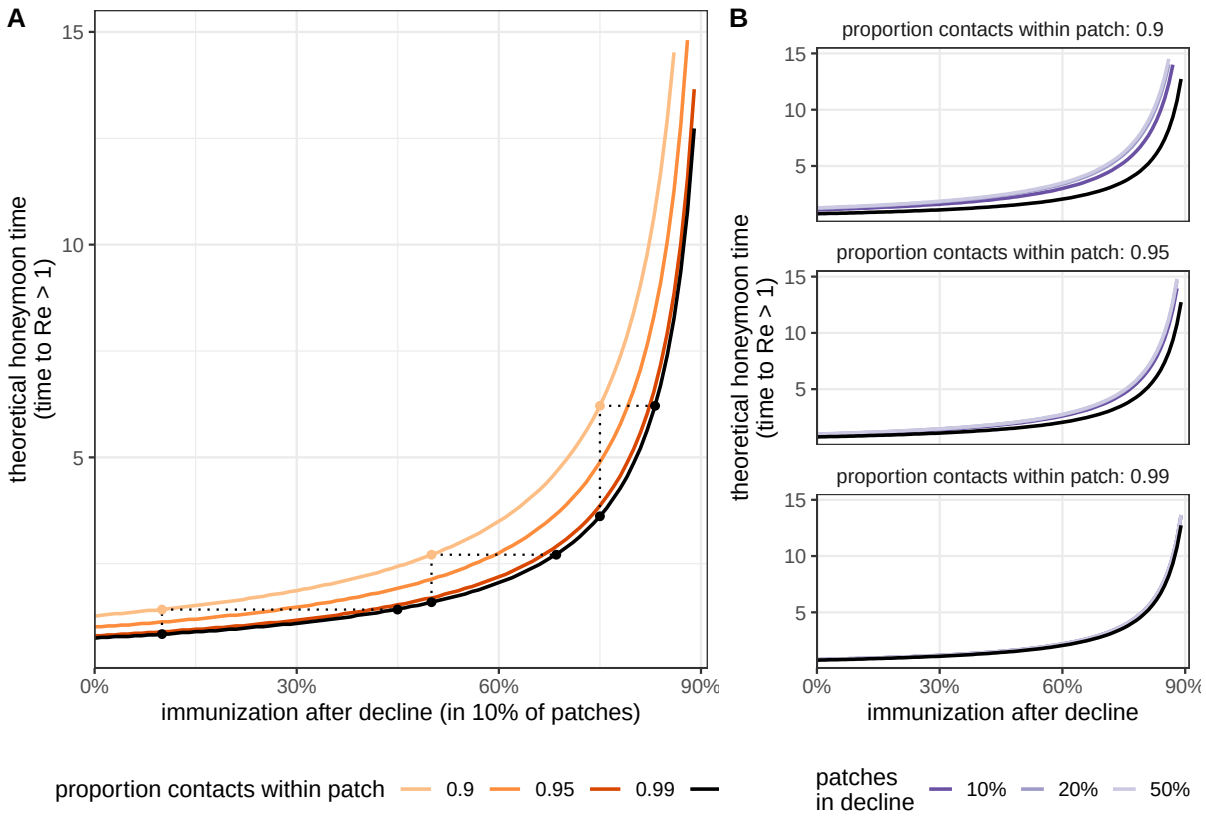



Figure S5: Role of patch connectivity in theoretical honeymoon time. (A) Theoretical honeymoon time, computed as time until $\mathcal{R}_{\text{eff}} > 1$, by post-decline effective immunization coverage when drops occur in 10% of patches among an equally connected network. Line color shows different connectivity scenarios, modeled by the proportion of contacts that occur within the home patch, and the points show equivalence between a model with no connectivity (i.e., 100% of contacts occuring within the home patch) and a model where 90% of contacts occur wtihin the home patch. (B) Sensitivity of results to the percent of patches in which effective immunization coverage declines compared to the no connectivity scenario (black). Results are shown for a population with $\mu = 1/80$ years$^{-1}$, a pathogen with $\mathcal{R}_0 = 17$, and pre-decline effective immunization coverage of 95%.

# References

[1] Diekmann O, Heesterbeek JAP, Metz JAJ. On the definition and the computation of the basic reproduction ratio R 0 in models for infectious diseases in heterogeneous populations. Journal of Mathematical Biology. 1990;28(4). doi:10.1007/bf00178324.

[2] Bjørnstad ON. Epidemics: Models and Data using R. Use R!. Springer International Publishing; 2018. doi:10.1007/978-3-319-97487-3.